

Problem F

Escape Route

Time limit: 2 seconds

Problem Description

In a tourist archipelago consisting of n islands, there are m bidirectional bridges connecting them. The islands are numbered $1, 2, \dots, n$. The i -th bridge connects islands u_i and v_i , and it takes t_i minutes to cross. Furthermore, each bridge has a height h_i , representing its height above sea level in centimeters.

An unprecedented natural disaster has struck the archipelago, causing the sea level to rise at a rate of 1 centimeter per minute. A bridge i becomes unusable exactly h_i minutes after the disaster begins. To ensure safety, a vehicle can only cross bridge i if the entire traversal completes before the bridge is submerged. Specifically, if a vehicle starts crossing bridge i at time T , it must arrive at the other end by time $T + t_i$, and this arrival time must satisfy $T + t_i \leq h_i$.

Island 1 serves as the evacuation center. Residents on each island are intelligent; they will choose a route that allows them to reach island 1 safely before any bridge they cross becomes submerged. If no such path exists, they will remain on their current island to await rescue.

Given this scenario, for each island k (where $k \in \{1, \dots, n\}$), determine if the residents on island k can safely reach island 1.

Input Format

The first line contains two integers n and m , representing the number of islands and the number of bridges, respectively.

Followed by m lines, the i -th of which contains four integers u_i, v_i, t_i, h_i , representing a bridge connecting islands u_i and v_i with travel time t_i and height h_i .

Output Format

Output a single string of length n . The i -th character of the string should be '1' if the residents of island i can safely reach island 1, and '0' otherwise.

Technical Specification

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 3 \times 10^5$
- $1 \leq u_i, v_i \leq n, u_i \neq v_i$
- $1 \leq t_i \leq h_i \leq 10^9$

Sample Input 1

```
3 2
1 3 2 4
2 3 3 5
```

Sample Output 1

```
1 0 1
```

Explanation for Sample 1: Residents of island 1 are already at island 1. Residents of island 3 can safely cross the first bridge and arrive at island 1 at time 2. However, residents of island 2 must first cross the second bridge to reach island 3 at time 3; if they then attempt to cross the first bridge, they will get stuck in the middle at time 4, exactly when the bridge becomes unusable since $h_1 = 4$. Therefore, the residents of island 2 cannot safely reach island 1.

Sample Input 2

```
5 6
1 2 4 4
1 4 8 13
2 3 1 3
2 4 3 7
3 5 2 2
4 5 7 8
```

Sample Output 2

```
1 1 1 1 0
```